MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548012

(COUNTER B, RANGE RATE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range rate counter B as used in the Range Computer CP-888/VPS-2, part of the Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548012	Counter B, Range Rate

PACKAGING DATA SHEET

10548012	Packaging of Counter B, Range Rate

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548012 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.5 through 3.5.8 logic input, output levels shall be as follows:

Logic 0 — 0.0 ±0.5 volt direct current (Vdc) level
Logic 1 — +3.5 ±1.0 Vdc level

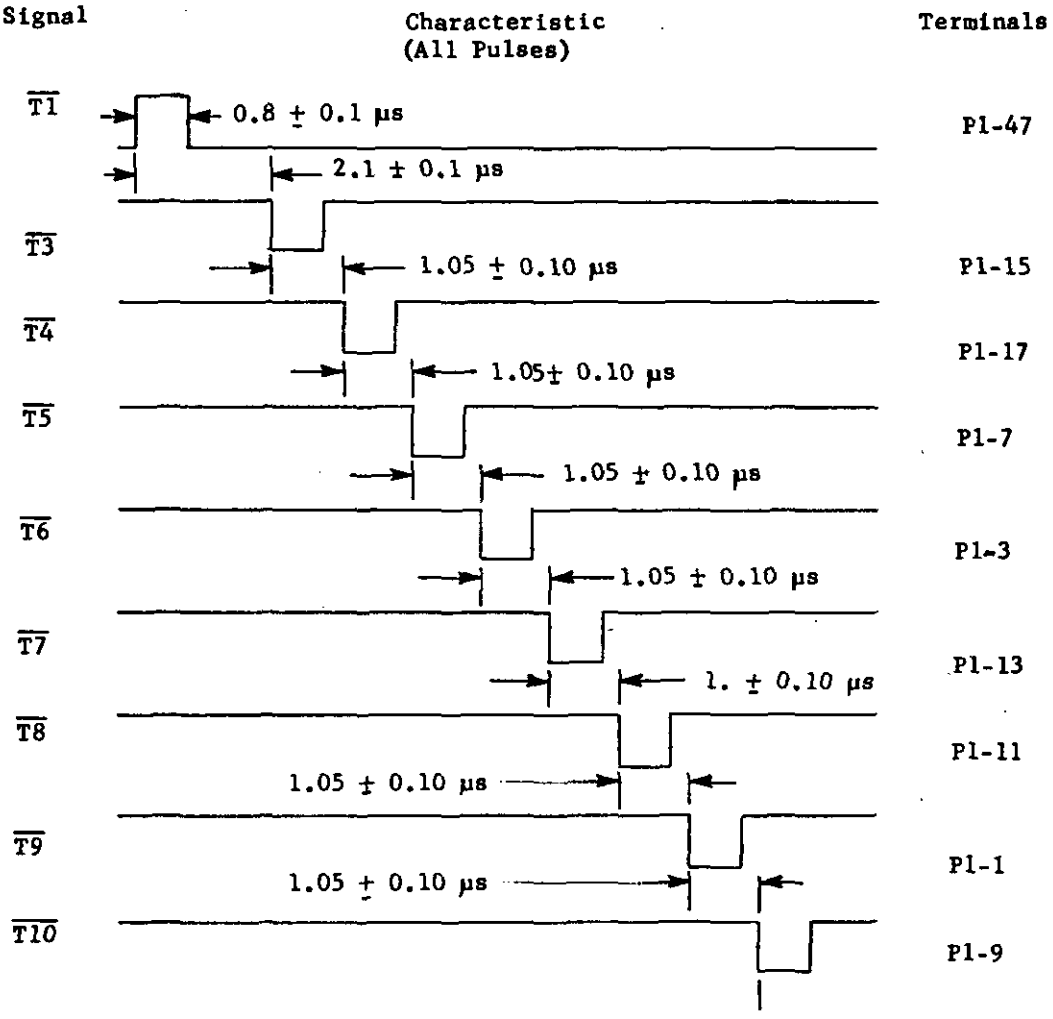
3.5.1 Input timing signals. - Timing input signals to the range rate counter B shall be in parallel form as shown in Figure 1. A positive pulse shall be a change in dc level from 0.0 ±0.5 volts (V) to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.2 Input power. - The input power applied between terminals P1-30, P1-31 and P1-32 (+5 V) and terminals P1-24, P1-25 and P1-26 (+5 V return) shall be 5.0 ±0.3 Vdc at 200 milliamperes (mA), regulated; with an alternating current (ac) ripple less than 50 millivolt (mV) peak-to-peak.

3.5.3 T3 through T10 serial output. - Timing pulses T3 through T10 shall be present at P1-8 in serial form. These pulses shall be complementary to, and have the same characteristics as $\overline{T3}$ through $\overline{T10}$ as specified in Figure 1.

3.5.4 To precount output. - Selectable precount pulse (s) shall be provided at P1-5 by jumpering any of the terminals E1 through E8 to any of the terminals E9 through E16, on the range rate counter B board. The precount pulse (s) selected shall have the same characteristics as specified in 3.5.3.

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All pulses at 20 kHz PRF.

FIGURE 1. INPUT TIMING SIGNALS.

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3.5.5 $\dot{R}$ serial output. - With input logic levels applied sequentially as specified in Table I, the $\dot{R}$ serial output signal at P1-20 shall be as specified in Table I. The output pulses shall have the same characteristics as specified in 3.5.3.

TABLE I. $\dot{R}$ SERIAL OUTPUT

	INPUT TERMINALS								
STEP	12	10	21	19	23	36	22	45	OUTPUT SIGNAL
1	1	1	1	1	1	1	1	1	T3 through T10
2	0	1	1	1	1	1	1	1	T4 through T10
3	0	0	1	1	1	1	1	11	T5 through T10
4	0	0	0	1	1	1	1	1	T6 through T10
5	0	0	0	0	1	1	1	1	T7 through T10
6	0	0	0	0	0	1	1	1	T8 through T10
7	0	0	0	0	0	0	1	1	T9 and T10
8	0	0	0	0	0	0	0	1	T10
9	0	0	0	0	0	0	0	0	Logic 0

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3.5.6 $+\overline{\dot{R}_O}$ output. - With input logic levels applied sequentially as specified in Table II, the $+\overline{\dot{R}_O}$ output signal at P1-18 shall be as specified in Table II.

TABLE II. $\overline{\dot{R}_O}$ OUTPUT

	INPUT TERMINALS					
STEP	19	23	36	22	45	OUTPUT SIGNAL
1	0	0	0	0	0	Logic 1
2	1	0	0	0	0	Logic 1
3	1	1	0	0	0	Logic 1
4	1	1	1	0	0	Logic 1
5	1	1	1	1	0	Logic 1
6	1	1	1	1	1	Logic 0

3.5.7 $-\overline{\dot{R}_O}$ output. - With input logic levels applied sequentially as specified in Table III, the $-\overline{\dot{R}_O}$ output signal at P1-34 shall be as specified in Table III.

TABLE III. $-\overline{\dot{R}_O}$ OUTPUT

	INPUT TERMINALS					
STEP	33	35	38	40	48	OUTPUT SIGNAL
1	0	0	0	0	0	Logic 1
2	1	0	0	0	0	Logic 1
3	1	1	0	0	0	Logic 1
4	1	1	1	0	0	Logic 1
5	1	1	1	1	0	Logic 1
6	1	1	1	1	1	Logic 0

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3.5.8 D/A output. - With timing pulse T1 applied to P1-47 and alternate logic levels of 0 and 1 sequentially applied as specified in Table IV, the D/A output signal shall be as specified in Table IV. Pulses shall have the same characteristics as specified in Figure 1 for T1, except for polarity.

TABLE IV. D/A OUTPUT

STEP	INPUT TERMINAL	OUTPUT TERMINAL	OUTPUT SIGNAL CHARACTERISTICS WITH 1 ON INPUT	WITH 0 ON INPUT
1	45	42	1 pulse	0 level
2	40	46	0 pulse	1 level
3	36	44	1 pulse	0 level
4	35	4	0 pulse	1 level
5	33	6	0 pulse	1 level
6	37	2	1 pulse	0 level
7	56	52	0 pulse	1 level
8	50	49	1 pulse	0 level
9	50	51	0 pulse	1 level

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

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4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table V shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 200 mA.
2. Data generator (digital word generator)	Nine parallel channels, 10 selectable bits per channel, with 1/0 content of each bit independently programmed. Data output shall be of return to zero (RZ) with controllable pulse width of 0.8 ±0.1 μs at a 20 kHz PRF. Output pulse amplitude of 5 V into a 50-ohm load.
3. Oscilloscope	Dual channel: rise time of scope plus measuring probe, 10 ns or less. Sweep period: 50 ns/cm or greater. Input resistance: 8 megohms or greater. Input capacitance: 25 pF or less.
4. Test fixture	As specified in Figure 2; see Table VI for test fixture parts list.

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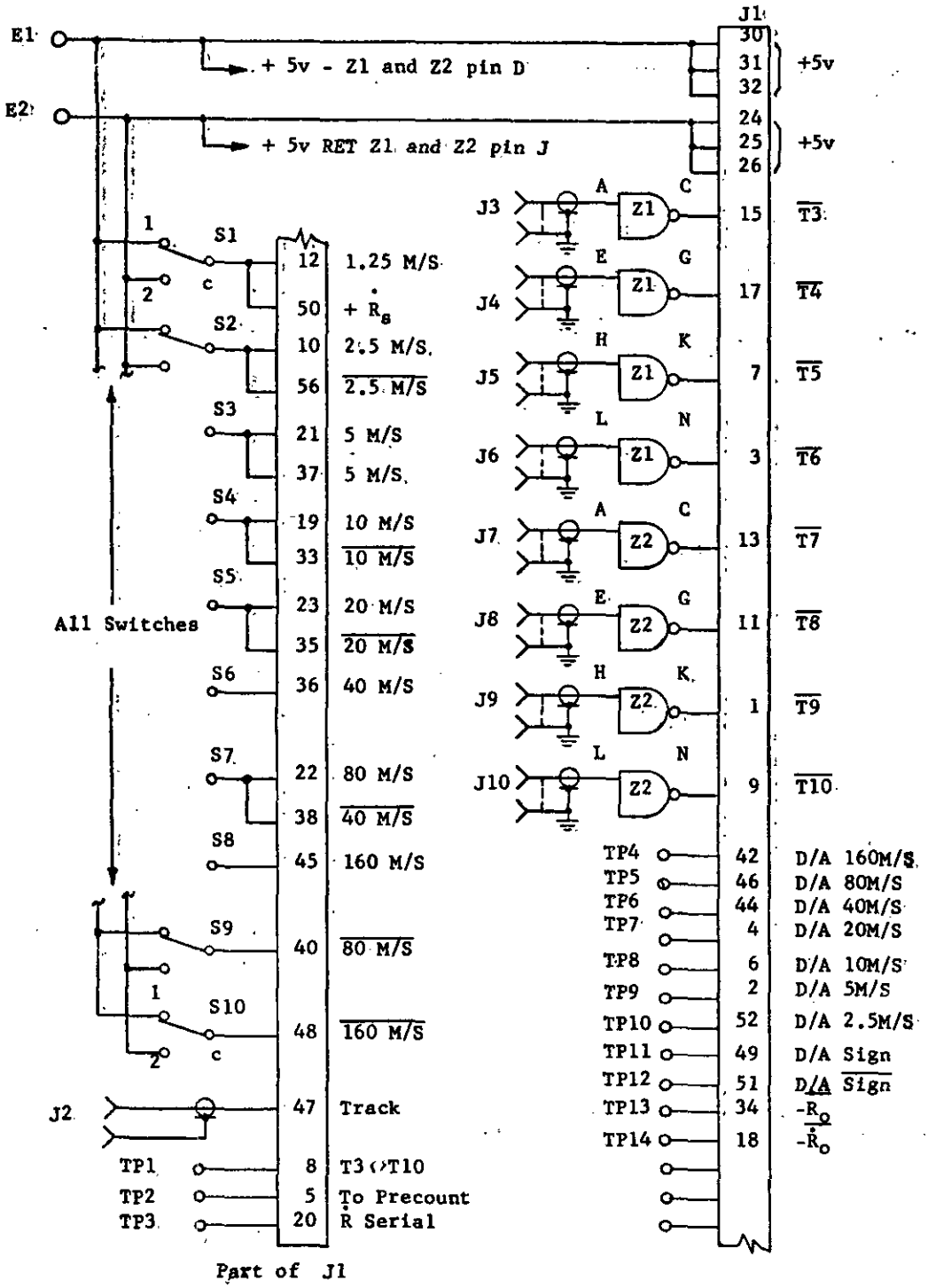


FIGURE 2. RANGE RATE COUNTER B TEST FIXTURE SETUP.

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TABLE VI. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2 thru J10	Connector, receptacle, UG-25B/U or equivalent.
S1 thru S10	Switch SPDT.
TP1 thru TP14	Tip jack, H. H. Smith 1501 or equivalent.
Z1 and Z2	Integrated circuit, SG-140-03 Sylvania or equivalent.

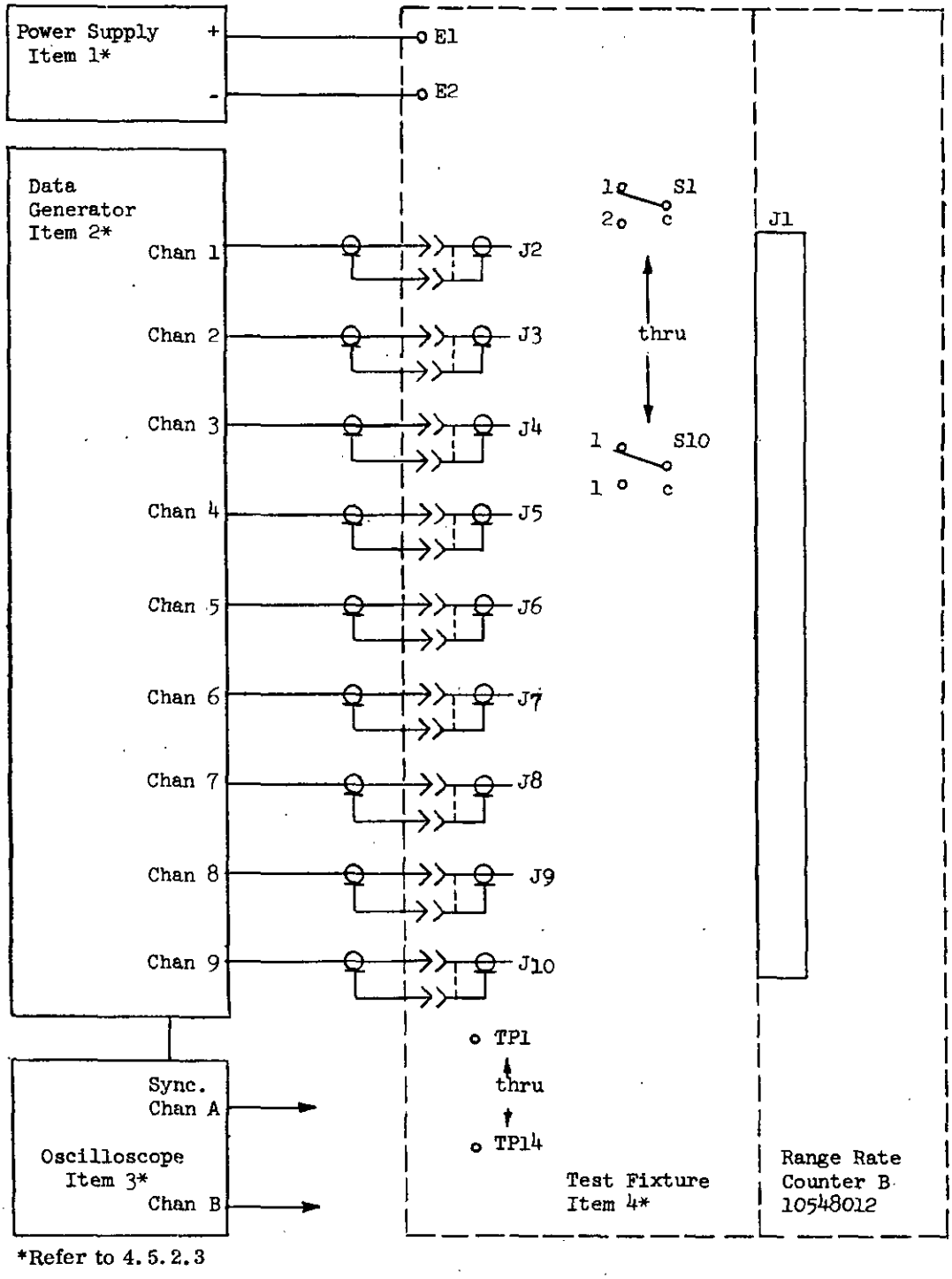
4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected range rate counter B shall be tested as follows: Set up equipment as shown in Figure 3. Use oscilloscope, item 3, for observation of input and output signals specified in the following paragraphs. When specified, a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a 3.5 ±1.0 Vdc level. Switches S1 through S10 on test fixture, item 4, are used to supply logic level input signals specified in Tables I through IV. Switches supply a logic 1 in position 1 and a logic 0 in position 2.

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*Refer to 4.5.2.3

FIGURE 3. TEST SETUP

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4.6.3.1 Input timing signals. - Using data generator, item 2, set up and apply timing signal T1 as shown in Figure 1 to J2 on test fixture, item 4. Apply timing signals T3 through T10 to J3 through J10 on the test fixture. Timing signals T3 through T10 shall have the same characteristics as $\overline{T3}$ through $\overline{T10}$ shown in Figure 1 except they shall be complementary to Figure 1.

4.6.3.2 Input power. - Using item 1, apply input power as specified in 3.5.2.

4.6.3.3 T3 through T10 serial outputs. - At TP1 on test fixture, item 4, observe that timing pulses T3 through T10 are present. The pulses shall be as specified in 3.5.3.

4.6.3.4 T_O precount output. - At TP2 on the test fixture, item 4, observe precount pulse(s) as selected by jumpering any of the terminals E1 through E8 to any of the terminals E9 through E16 on the range rate counter B board. The selected precount pulse(s) shall have the same characteristics as specified in 3.5.3.

4.6.3.5 R serial output. - Using switches S1 through S8 to apply logic levels to P1-12, P1-10, P1-21, P1-19, P1-23, P1-36, P1-22, and P1-45 respectively, sequentially perform the steps as specified in Table I. At TP3 observe output signal as specified in Table I with characteristics specified in 3.5.3.

4.6.3.6 $+\overline{\dot{R}_O}$ output. - Using switches S4 through S8 to apply logic levels to P1-19, P1-23, P1-36, P1-22, and P1-45 respectively, sequentially perform the steps as specified in Table II. At TP14 observe the output signal as specified in Table II.

4.6.3.7 $-\overline{\dot{R}_O}$ output. - Using switches S4, S5, S7, S9, and S10 to apply logic levels to P1-33, P1-35, P1-38, P1-40, and P1-48 respectively, perform the steps as specified in Table III sequentially. At TP13 observe the output signal as specified in Table III.

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4.6.3.8 D/A output. - Sequentially use switches specified in Table VII to alternately apply logic 0 and 1 levels as specified in Table VII. At test points listed in Table VII observe the signals as specified in Table VII. The pulses observed shall be as specified in 3.5.8.

TABLE VII. - SIGNAL CHARACTERISTICS

TEST POINT	SWITCH	SIGNAL CHARACTERISTICS	
		LOGIC 1	LOGIC 0
TP4	8	1 pulse	0 level
TP5	9	0 pulse	1 level
TP6	6	1 pulse	0 level
TP7	5	0 pulse	1 level
TP8	4	0 pulse	1 level
TP9	3	1 pulse	0 level
TP10	2	0 pulse	1 level
TP11	1	1 pulse	0 level
TP12	1	0 pulse	1 level

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548012. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The range rate counter B is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The range rate counter B converts range rate data, supplied in parallel form to serial form; provides a means to select precount pulses any time from T3 through T10; and detects a fully loaded or fully empty condition of range counter A.

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6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

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Preparing Activity:

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Project No. 1285-A247

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO (If "yes", in what way?)

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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